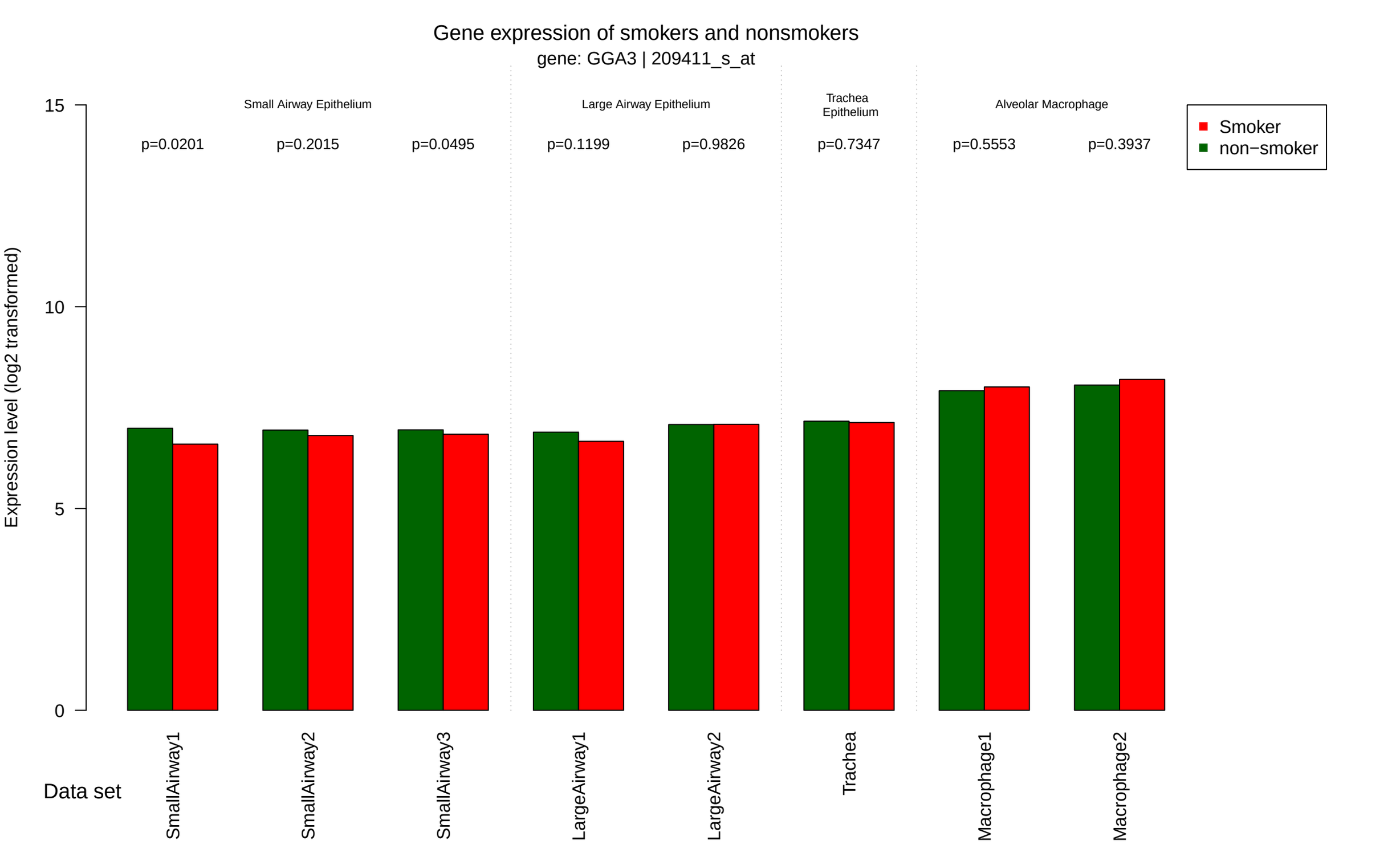
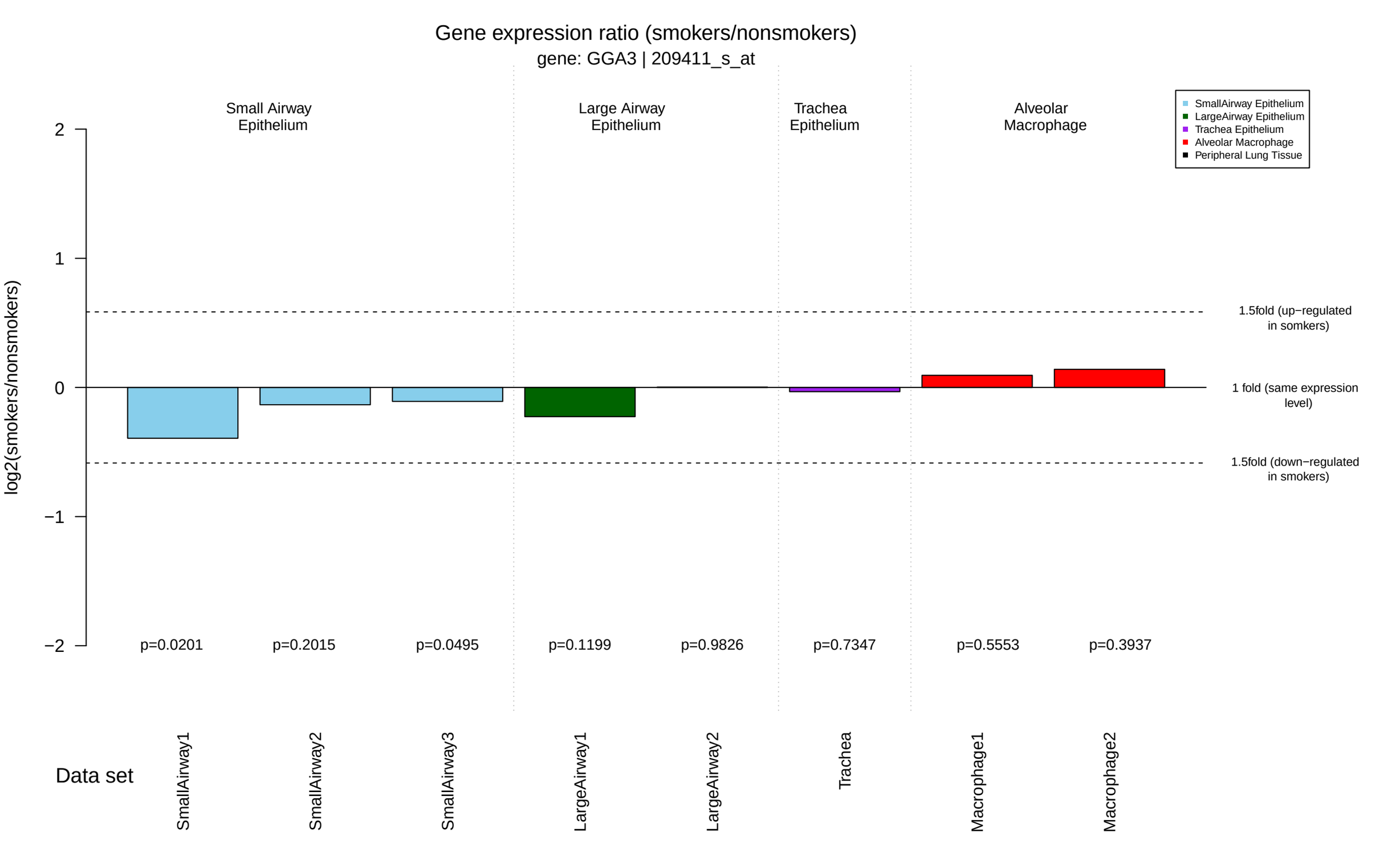
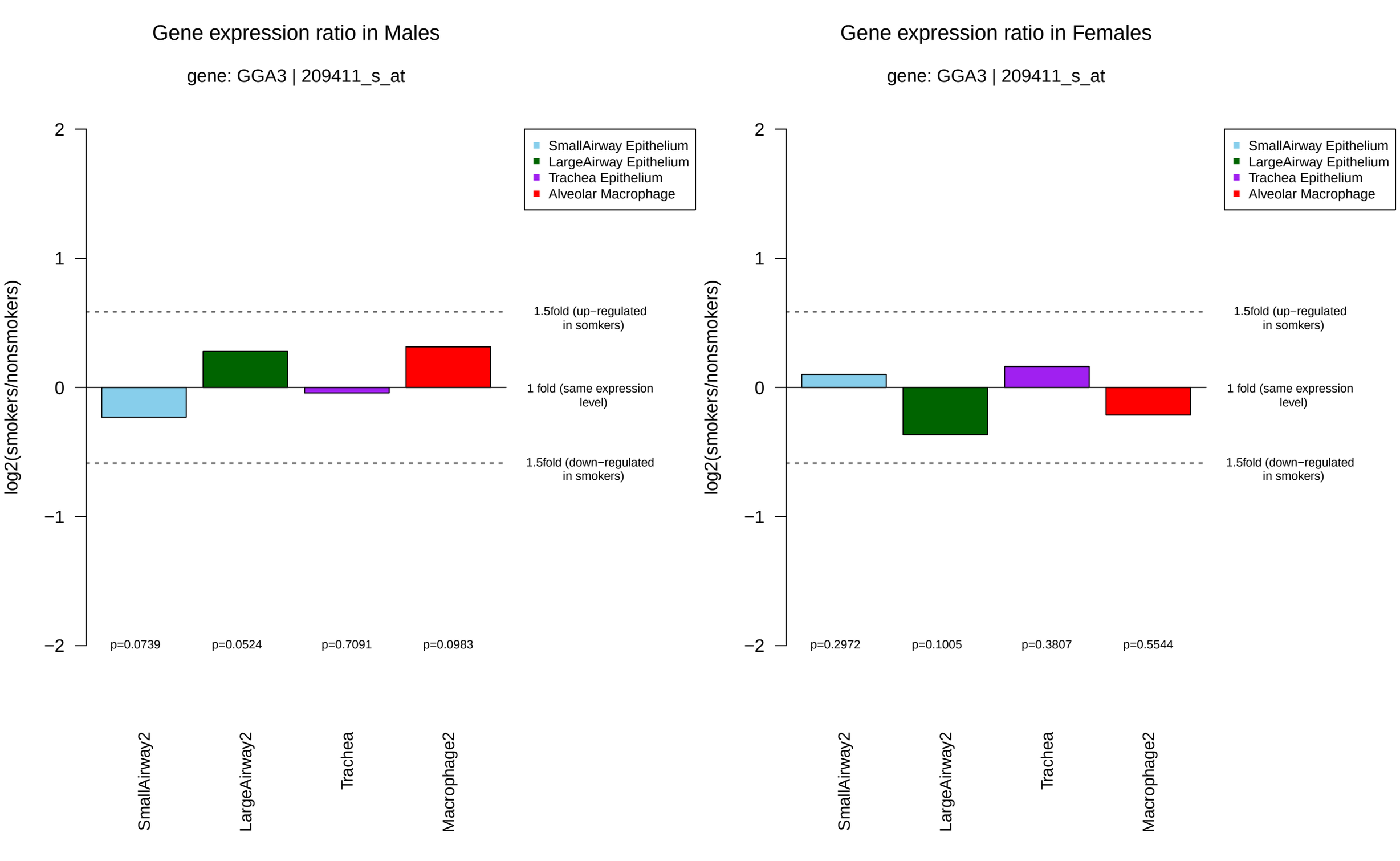
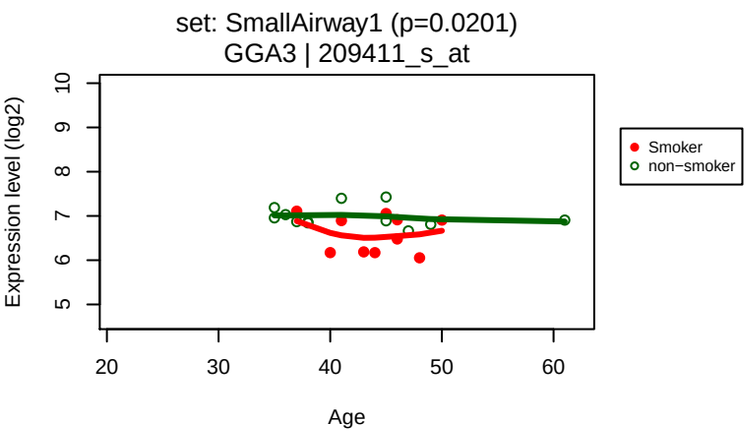
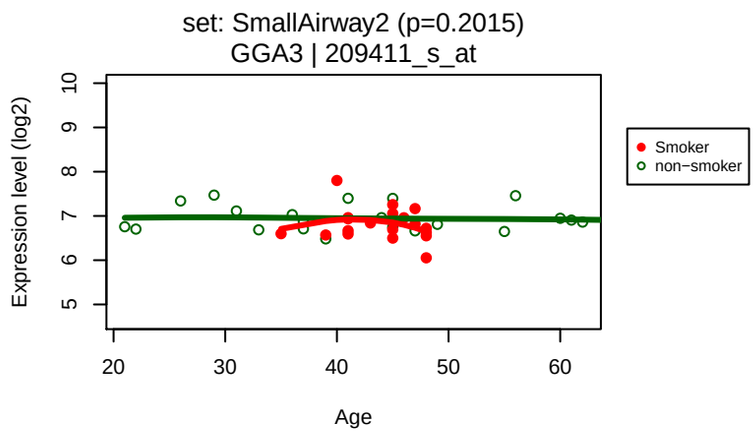
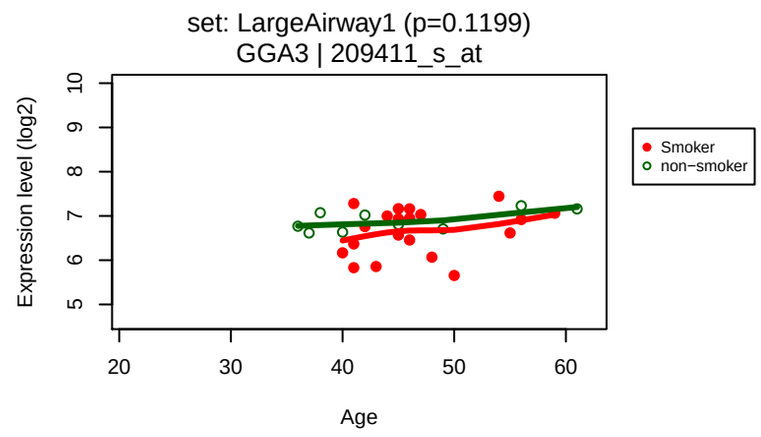

Gene expression of smokers and nonsmokers
gene: GGA3 | 209411_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0201
p=0.2015
p=0.0495
p=0.1199
p=0.9826
p=0.7347
p=0.5553
p=0.3937
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

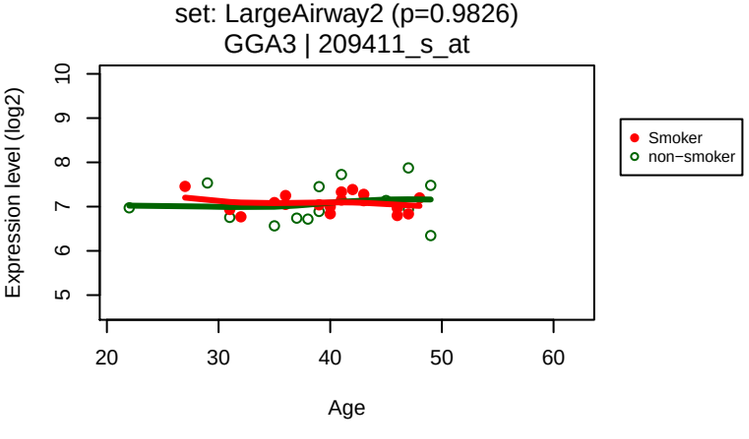

Gene expression ratio (smokers/nonsmokers)
gene: GGA3 | 209411_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0201
p=0.2015
p=0.0495
p=0.1199
p=0.9826
p=0.7347
p=0.5553
p=0.3937
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

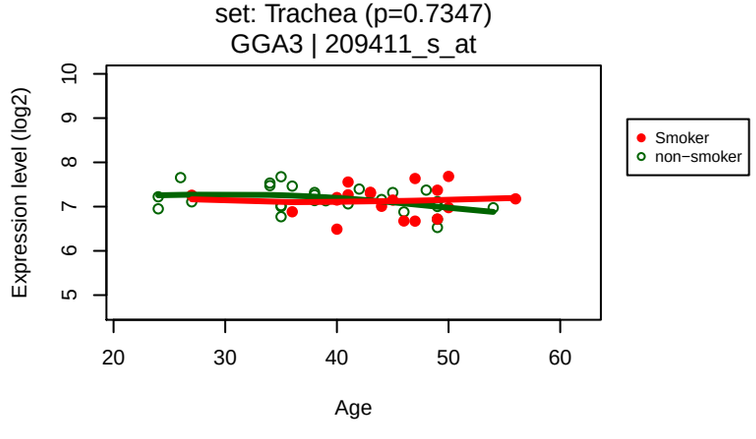

Gene expression ratio in Males
gene: GGA3 | 209411_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0739
p=0.0524
p=0.7091
p=0.0983
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GGA3 | 209411_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2972
p=0.1005
p=0.3807
p=0.5544
SmallAirway2
LargeAirway2
Trachea
Macrophage2

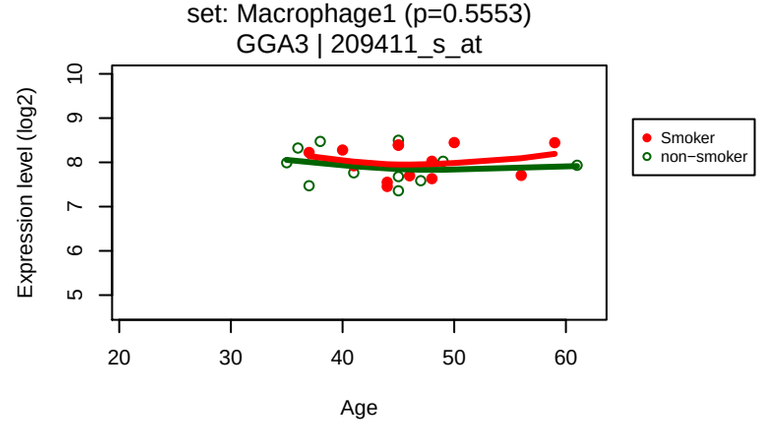

set: SmallAirway1 (p=0.0201)
GGA3 | 209411_s_at
Expression level (log2)
Age
Smoker
non−smoker

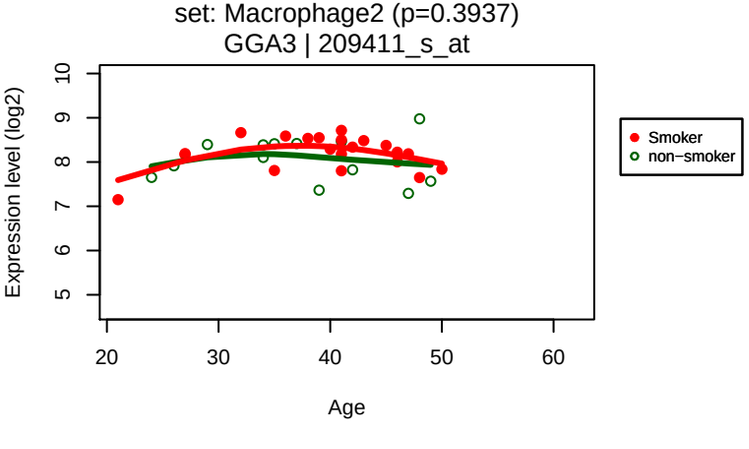

set: SmallAirway2 (p=0.2015)
GGA3 | 209411_s_at
Expression level (log2)
Age
Smoker
non−smoker

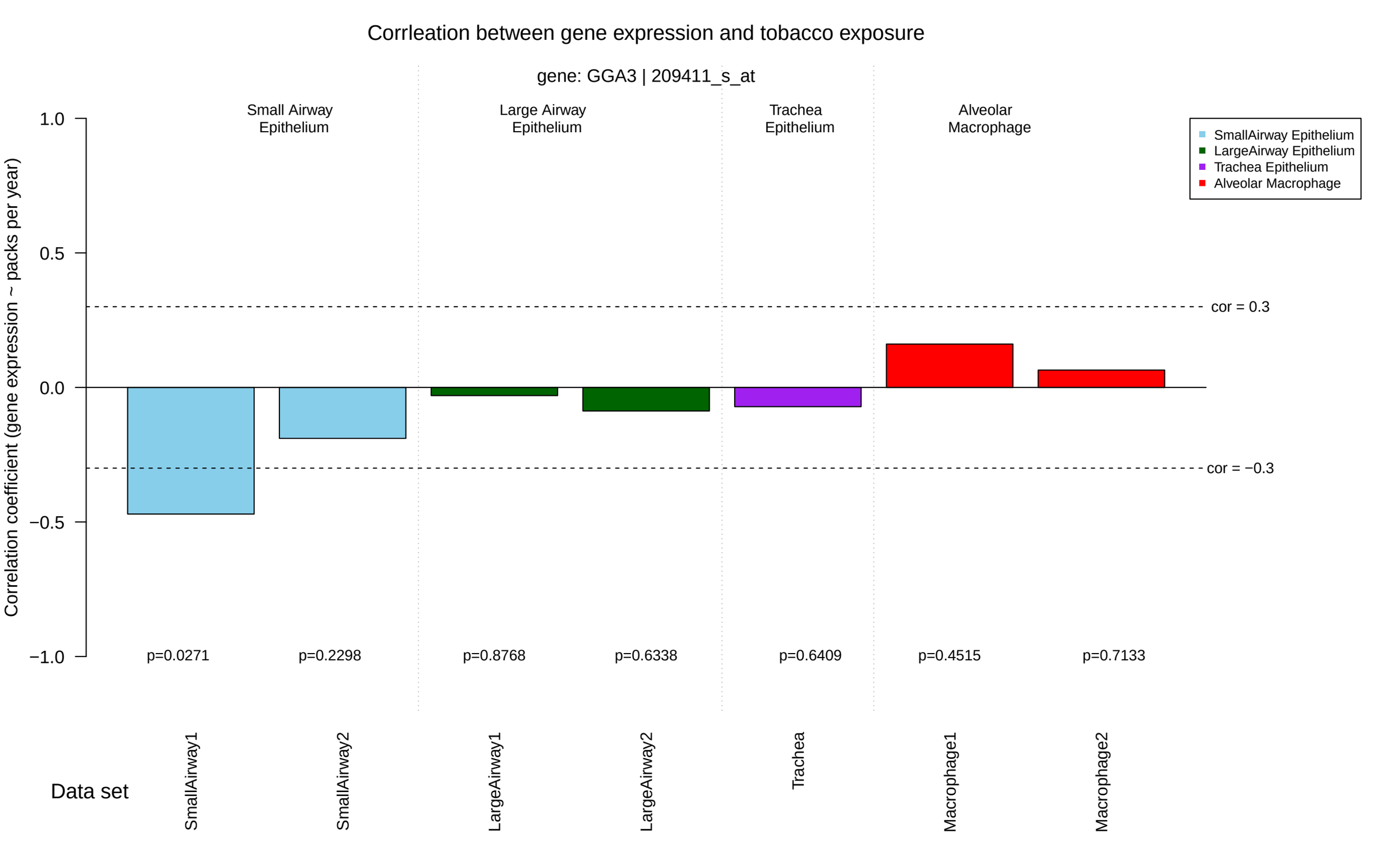

set: LargeAirway1 (p=0.1199)
GGA3 | 209411_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9826)
GGA3 | 209411_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7347)
GGA3 | 209411_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5553)
GGA3 | 209411_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3937)
GGA3 | 209411_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GGA3 | 209411_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0271
p=0.2298
p=0.8768
p=0.6338
p=0.6409
p=0.4515
p=0.7133
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2